

A Multifaceted Analysis of Intelligent Vehicle Route Optimization

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Abstract—In the constantly changing realm of logistics and transportation management, the incorporation of Information and Communication Technology (ICT) has catalyzed transformative paradigm shifts in the approach and resolution of Vehicle Route Optimization (VRO). The current scientometric research paper embarks on a comprehensive exploration of the scholarly endeavors acquired from the Scopus database in the realm of ICT-assisted vehicle route optimization, spanning 2014–2023. The scientometric implications of the article encompass several pivotal dimensions, including publication patterns, (author, country, institution) co-authorship, geographical distribution, Document Co-citation network Analysis (DCA) and top articles based on betweenness centrality corresponding to each opted category of the current knowledge domain. A meticulous examination of the analyses revealed a significant research impact in the pervasive computing and communication technology categories. The co-authorship analysis presenting the interconnectedness of collaborative efforts across countries, authors, and institutions highlights the authors and universities of China and the United States as dominant players in the domain. The DCA elucidates research themes, including intelligent transportation systems, unmanned aerial vehicle-based wireless sensor networks, electric vehicle-based sustainable VRO, and vehicular ad-hoc networks. These themes underscore the current research trajectories within the field. Notably, quantum computing and blockchain emerged as prominent technologies. Overall, the study unveils the transformative impact of ICT on VRO, highlighting the key themes, future research directions and a collaborative research community poised for substantial innovation in this area of research.

Index Terms—Dynamic vehicle routing problem (DVRP), vehicle to everything (V2X), vehicular ad-hoc networks (VANET), Internet of Vehicles (IoV), CiteSpace.

I. INTRODUCTION

IN THE 21st century, the world has witnessed an unprecedented boom in technology, fundamentally reshaping industries and driving advancements in every facet of human life. Among the myriad applications of modern technology, the utilization of ICT has emerged as a transformative force in optimizing various processes, enhancing productivity, and promoting sustainability. One area where the fusion of ICT has demonstrated remarkable potential is in Vehicle Route

Optimization (VRO) – a critical aspect of transportation and logistics management that profoundly impacts businesses, economies, and the environment. From local deliveries to global supply chains, efficient vehicle route planning and optimization have become essential to ensure the smooth functioning of economies and reduce the adverse effects of transportation on the environment. With the rise of ICT and the advent of powerful computational algorithms, the transportation industry is undergoing a seismic shift towards dynamic and intelligent route optimization systems. ICT incorporates Artificial Intelligence (AI), big data analytics, mobile technology, communication systems, global positioning systems, and various VRO algorithms that enhance the overall operational efficiency of the vehicle navigation system.

Integration of ICT for VRO holds profound implications, including accurate delivery estimations, reduced traffic congestion, reduced fuel consumption and greenhouse gas emissions, and intelligent transportation solutions. AI, Machine Learning (ML), and Deep Learning (DL) techniques are significantly employed for making real-time traffic flow prediction, management, and intelligent transportation systems [1], [2]. Furthermore, big data analytics have improved the logistics operational efficiency and customers' experience because of the surge in extensive traffic data originating from transactions, social media tools, and sensor networks [3], [4]. The location-based services integrate data by leveraging technologies such as global positioning system devices, traffic sensors, closed circuit television cameras, Internet of Things (IoT) sensors, and geographical databases to create a comprehensive and real-time spatial database. The lack of efficient traffic data collection infrastructure hinders the usage of readily available vehicle routing systems [5]. Moreover, in real-world scenarios, finding the optimal vehicle route poses formidable challenges due to multiple constraints such as time window, capacity, pickup and delivery, and precedence, rendering it an NP-hard problem. For solving the complex VRO problems, an array of potent optimization algorithms encompass exact, heuristic, metaheuristic, and learning-based approaches [6], [7]. These algorithms offer a comprehensive toolkit to address multifaceted and small-scale VRO problems efficiently. However, in the noisy intermediate-scale quantum era, the advent of quantum computing techniques with their remarkable parallel processing capabilities has sparked widespread interest in quantum-inspired optimization algorithms. The meteoric rise in popularity of these cutting-edge algorithms stems from their exceptional ability to swiftly generate solutions for intricate

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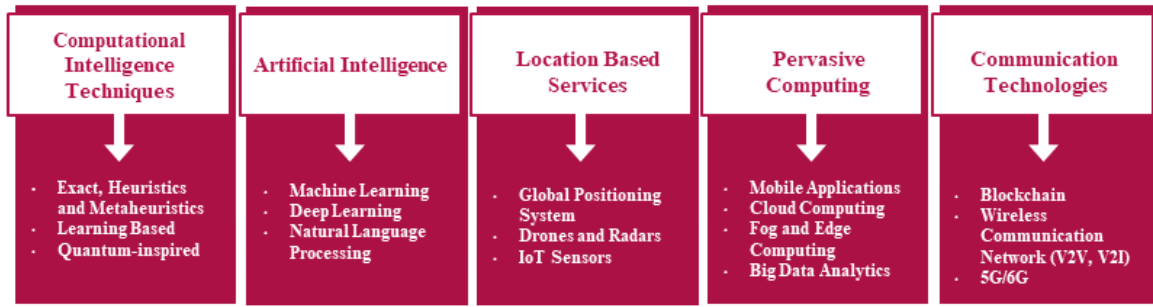


Fig. 1. ICT categories.

and scalable VRO challenges, a feat that conventional techniques struggle to achieve within feasible timeframes [8]. In a nutshell, various computational intelligence techniques can integrate ICTs to provide optimal solutions in transportation and operational research.

The advent of mobile computing and telecommunication technologies, including telematics, mobile apps, and communication systems, has created substantial prospects in various domains:

1) Transportation agencies now have valuable tools at their disposal for activities like gathering travel data, disseminating travel information, promoting ride-sharing and carpooling, and enhancing traffic safety.

2) Public travelers can harness these advancements for purposes of planning their routes, organizing ride-shares or carpools, finding parking spaces, ensuring traffic safety, and accessing real-time travel information.

Extensive use of smart mobile applications can be highly advantageous, especially for automobile mode, as it helps decrease travel time, costs, and vehicle emissions [9]. The advancements made by ICT for VRO have led to the pressing need for comprehensive research to assess the developing paths and significant research contributions to bolster the performance of current transportation systems.

Subsequently, the vast ICTs are broadly divided into five parts by deeply analyzing the existing literature about the technologies included in ICT [10], [11], [12], [13], [14]. Based on the recommendations given by the researchers in that literature these technologies have been grouped into five categories as illustrated in Fig. 1, including, Computational Intelligence Techniques (CIT), AI, Location Based Services (LBS), Pervasive Computing (PC), and Communication Technologies (CT). In each category, a wide range of relevant technologies have been included with the aim to broadly cover the whole technological landscape of VRO. Furthermore, towards the objective, a comprehensive scientometric analysis and critical literature review of ICT-assisted VRO literature is conducted to discern the development landscape of the knowledge domain. The study makes the following contributions to the current knowledge landscape in this domain:

1) The study presents a comprehensive overview of how the ICT-assisted VRO research domain has evolved and adapted over time.

2) It maps the global network of research, pinpointing leading hubs of innovation and centers of knowledge exchange that drive advancements in ICT-assisted VRO.

3) The research enables a detailed exploration of the current knowledge landscape, facilitating a deeper understanding of the current advancements and research trends in this knowledge domain.

4) By examining pivotal works from the past decade, the study uncovers interdisciplinary connections, to present broader implications of the research.

A. Literature Review

The current research explores the significant articles published in the ICT-assisted VRO research domain. Boukerche et al. [15] performed a in-depth survey of using AI technologies for predicting the traffic flow in intelligent transportation systems. Notably, the paper thoroughly discussed two primary prediction approaches, statistical models and ML-based prediction models, along with the prominent algorithms associated with each approach. Tong et al. [16] presented an exhaustive survey of the literature utilizing AI technology to encounter challenges in V2X system research. The study inferred that AI-driven algorithms have shown better performance compared to conventional counterparts in terms of elevating the comfort, safety, and efficiency of driving in a V2X environment. Cheng et al. [17] provided a thorough examination of the capabilities of big data technologies to enhance VANETs. Additionally, the authors have developed a framework aimed at effectively leveraging big data in VANETs to enhance their overall performance. Banerjee et al. [18] performed an analysis of traffic management and forecasting by utilizing IoT technology. The article discussed the development of embedded computing systems using IoT and the trends in the industry to develop IoT platforms for efficient traffic management. Osaba et al. [19] reviewed the research carried out on the integration of quantum computing and Vehicle Routing Problems (VRPs), and the analysis shows an inclination towards practical research due to the advancements in technology. Eksioglu et al. [20] proposed a broad classification of VRP and delineated VRP's facets according to the proposed taxonomy. This valuable contribution not only offers a systematic literature review but also highlights future research directions in this domain. The authors proposed

a comprehensive survey of DVRPs primarily focusing on applications and solution methods. The findings reveal that advancements in ICT have the potential to unlock exciting new horizons within this domain [21], [22].

This literature review reveals that numerous researches have showcased the significance of ICTs in VRO. However, there is a notable lack of study discerning the current advancements in this knowledge landscape that could guide future research and promote collaboration among researchers. To address this gap, the subsequent research questions have been framed:

- 1) What are the annual publication trends in the ICT-assisted VRO research domain?
- 2) What are the collaboration patterns of authors, institutes and countries in the contemporary knowledge landscape of this field?
- 3) What are the prevailing research themes and various ICTs effectively used in the last decade?
- 4) Which are the significant publications that have shaped the field of ICT-assisted VRO?
- 5) What are the future research directions, algorithm selection and research implications pertinent to the current state of knowledge in ICT-assisted VRO?

Paper Organization: The remaining portion of the article is organized into five sections, where, Section II delineates the methodology of the proposed research, which describes the research approach, data collection strategy and analytical tool selection procedure. Section III is an analyses section, which includes publication growth analysis, co-authorship network analysis (country, author, institution), DCA, and top articles based on Betweenness Centrality (BC). Section IV elucidates the discussion, algorithm selection for VRO, future research directions and research implications. Section V outlines the conclusion and limitations of the study.

Note: Figures and Tables labeled with “S” are included in the supplementary file.

II. RESEARCH FRAMEWORK

A. Research Approach

Traditional manual/systematic literature review methods, which are often qualitative and introduce subjective bias [23]. The scientometric approach provides a rigorous analytical framework and visualization techniques to identify and understand trends, and developing paths in a field. Moreover, unlike bibliometric, which primarily focuses on publication and citation counts, scientometric analysis leverages bibliometric data including citation counts, references, authors, abstract, keywords, and journal detail, to uncover collaboration patterns, research trends, key entities (references, authors, research areas) and research hotspots within the field. Chen et al. [24] described scientometry as the analysis of published literature using various bibliometric methods. The purpose is to map the structure and development of a specific field by examining a curated collection of high-quality academic works. Scientometrics has become a widely used approach to delineate the developmental path of a domain for researchers, due to its significance and broad applicability [25]. Given the large and growing body of literature on ICT-assisted VRO,

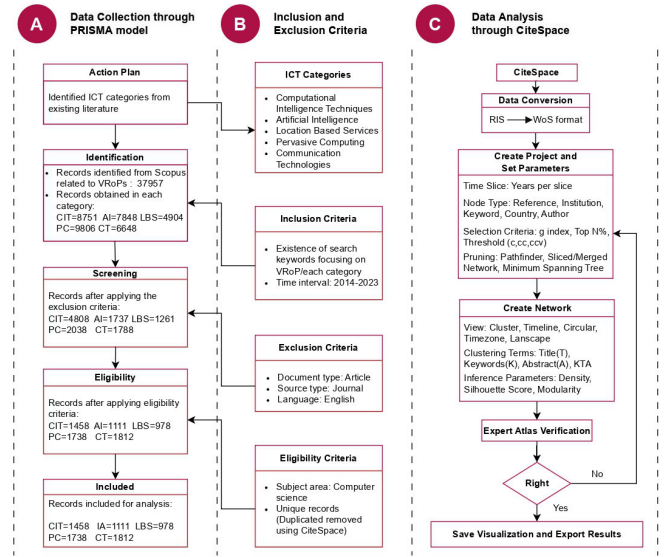


Fig. 2. Research methodology.

the scientometric approach offers the most robust framework for analyzing the research landscape and informing future directions, making it the optimal choice for the current study.

B. Tool Selection

Researchers have access to various data analysis and visualization tools for performing scientometric analysis, such as CiteSpace [24], Gephi [26], VOSviewer [27], SciMAT [28], and UCInet [29]. For this study, CiteSpace a Java-based software is utilized, as it provides various advanced functionalities for exploring, analyzing, and visualizing bibliometric data as compared to other visualization tools. Table I presents a comparison between CiteSpace, Gephi, VOSviewer, and UCInet based on various metrics. CiteSpace excels in identifying intellectual turning points, tracking the evolution of research topics over time, and detecting key literature through BC. Unlike VOSviewer or Gephi, it also supports temporal slicing and co-citation clustering, which are essential for analyzing the dynamic structure of scientific fields. Moreover, the ability of CiteSpace to map co-occurrence networks across multiple dimensions, such as countries, authors, keywords, institutions, and journals, makes it particularly suitable for comprehensive bibliometric exploration. Furthermore, the C part of Fig. 2 showcases the flow of analytics paths supported by CiteSpace for creating visualizations aiming at assessing research documents/authors/country/journals/institution/keywords/phrases/subject categories for delineating co-occurring patterns, co-citation patterns, research hotspots, intellectual turning points, and emerging research areas.

C. Data Collection

The current section presents the methodology adopted to curate the dataset from the Scopus database to perform a scientometric analysis of VRO literature in the five ICT categories including CIT, AI, LBS, PC and CT. The Scopus database is

TABLE I
COMPARISON OF VISUALIZATION TOOLS

Feature	CiteSpace	Gephi	VOSviewer	UCInet
Burst Detection	Yes	No	No	No
Centrality Metrics	Yes	No	No	No
Temporal Metrics and Time-series Analysis	Yes (Time Slicing, Timeline view)	No	No	No
Data Preprocessing and Data Cleaning	Yes	No	No	No
Cluster Label Extraction	Yes	No	No	No
Node Pruning	Yes	No	No	No
Visualization Views	Network, timeline, timezone, heatmap, circular, overlay, cluster, landscape	Graphs, plots	Network, overlay and density	Graphs, plots
Bibliometric Analysis	Co-authorship (keywords, institute, country), co-citation (references, journals, authors), co-occurring phrases, co-occurring subject categories analysis, dual-map overlay	No	Co-authorship, co-citation, co-occurrence, bibliographic coupling	No
Clustering Support	Mature functionality (GPT/LLM-generated)	Good for large datasets	Limited	Limited
Input/Output Formats	.txt, .csv, RIS, excel, .png, .svg, .html	.gexf, .csv, .svg, .png	RIS, Endnote, Refworks, .svg, .eps, .gif, .jpg, .png, .svg, .swf, .tiff	ASCII text files, dl, Raw
Platform Support	Java-based	Java-based, Open GL	Windows, Linux, Mac	Windows
Database Support	WoS, Scopus, Dimensions, S2AG/MAG, CNKI/WanFang, CSDC, CSSCI, PubMed	MySQL, PostgreSQL	WoS, Scopus, Dimensions, Lens	No

chosen in this study because it is an extensive, high-impact, and reputable corpus of scientific literature, covering a wide range of disciplines. Fig. 2 visualizes the steps carried out for data collection through Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) methodology and for creating science maps using CiteSpace visualization tool [30]. A widely adopted and efficient PRISMA methodology encompasses a systematic process characterized by four distinct phases: Identification, Screening, Eligibility, and Inclusion.

In the Identification phase, the records related to the VRO domain were tailored using the search string containing keywords related to VRO, as mentioned in set K_{VRO} of Eq. 1. This set of keywords was used on the (TITLE-ABS-KEY) search field of the Scopus database to get a cumulative count of 37957 records. Subsequently, to retrieve data for a specific category, a set of keywords (K_{Ci}) associated with category i was employed as a subquery in the Scopus database. This process yielded 8751, 7848, 4904, 9806, and 6648 records in the CIT, AI, LBS, PC, and CT categories, respectively, following the inclusion criteria.

In the screening phase, the dataset was refined to align the records as per the requirements and objectives of the study, while minimizing overlapping among the categories. To adhere the records to the requirements and objectives of the study the exclusion criteria mentioned in Fig. 2 was applied. Moreover, to reduce overlapping, redundant records within a category were removed by employing “AND NOT” constraint, applied to the keyword sets of all other categories, as specified in Eq. 1. The “AND NOT” constraint ensures that the keyword set to which it is applied does not appear in the targeted category.

During the eligibility phase, the CiteSpace tool was employed to effectively eliminate any duplicate records and further refined the dataset to focus exclusively on the Computer Science research area.

Consequently, the final dataset comprised 1458, 1111, 978, 1738, and 1812 papers about the CIT, AI, LBS, PC, and CT categories, respectively. The records are downloaded in Research Information System (RIS) file format with citation information, bibliographical information, abstract and keywords, funding details, and other information to perform analysis. These comprehensive datasets were subsequently

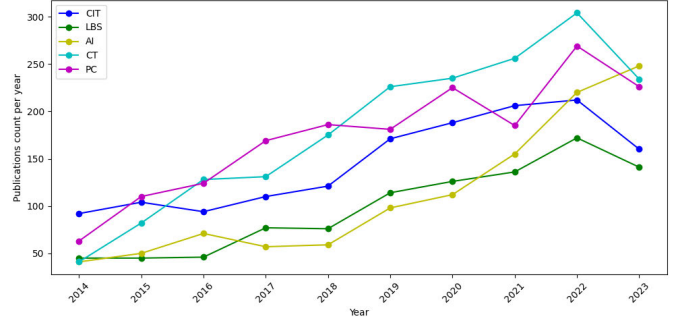


Fig. 3. Publications trends corresponding to each category.

analyzed using the CiteSpace tool to uncover research trends, collaborations, and thematic structures within each category. Specifically, the DCA and BC metric facilitated the identification of more than 100 highly influential publications. This systematic approach ensured a robust analysis of the intellectual landscape and the extraction of meaningful insights from the extensive collection of records.

$$SQ_i = \left[(K_{VRO}) \cap \{ (K_{Ci}) \cap - \left(\bigcup_{j=1, j \neq i}^n K_{Cj} \right) \} \right] \cap (EC) \quad (1)$$

where, K_{Ci} are the set of keywords corresponding to each category, EC is the set of exclusion criteria, n is the total number of categories which is five

$K_{VRO} = \{\text{vehicle route OR vehicle routing problem OR vehicle route optimization}\}$

$K_{CIT} = \{\text{quantum inspired OR metaheuristic OR exact solution OR heuristic}\},$

$K_{CAI} = \{\text{artificial intelligence OR deep learning OR neural network OR machine learning}\},$

$K_{CLBS} = \{\text{geographic information system OR global positioning system OR sensors OR drones OR radar}\},$

$K_{CPC} = \{\text{cloud computing OR edge computing OR mobile computing OR internet of things OR autonomous vehicle transport OR big data OR data mining}\},$

$K_{CCT} = \{\text{blockchain OR 5g OR 6g OR wireless communication network OR vehicle to vehicle OR vanet}\}.$

TABLE II
PUBLICATIONS ANALYSIS IN EACH CATEGORY

Category	TP	TP%	TC	ACPP
Computational Intelligence Techniques (CIT)	1458	20.54	32381	22
Artificial Intelligence (AI)	1111	15.65	18907	17
Location Based Services (LBS)	978	13.78	15303	15
Pervasive Computing (PC)	1738	24.48	41234	23
Communication Technologies (CT)	1812	25.53	37443	20

Note: TP = Total Publications, TP% = Percentage of Total Publications, TC = Total Citations, ACPP = Average Citations per Publication.

III. ANALYSES

A. Publication Growth

The temporal publication patterns provide insights into the overall publication trends, pace of development, and academic interest within the research field. Fig. 3 presents the year-wise publication growth corresponding to each ICT category in VRO domain. The data reveals a general upward trend in all categories, indicating the rise in research efforts in the current knowledge domain. Notably, a plateau around 2020-2021 is observed which is likely due to the COVID-19 pandemic, however, the recovery from 2022 onwards is evident, suggesting significant interest and accelerated research efforts. Furthermore, the graph shows that the publication rate in the PC and CT categories has observed a significant escalation throughout the decade, indicating the development and maturity of this category.

Table II shows the publications and citations trends, considering Total Publications count (TP), publications share (TP%), Total Citations (TC), and Average Citations Per Publication (ACPP) corresponding to each category. Specifically, TP gives a direct measure of how much research output exists for each category; TP% offers a relative view of how much each category contributes to the overall research landscape and provides the proportion of work done in each category as compared to others; TC indicates the influence and impact of the research as a higher citation count reflects that the work is being widely recognized and used by other researchers; and ACPP offers the influence of individual publication. Notably, the PC category has garnered the highest (41234) number of citations indicating high-quality and insightful scientific work. Moreover, the CT category leads with the highest TP and TP% (1812, 25.53%) indicating substantial research output and influence. Additionally, CT category closely follows the PC category in terms of TC count, suggesting a similar level of impact. In contrast, the LBS category has received a minimum TC (15303), indicating the requirement of researchers' efforts in this category.

B. Co-Authorship Network Analysis

Co-authorship analysis delineates the publications' status at the micro level. Co-authorship is a prominent and well-documented form of scientific collaboration, occurring at three distinct levels: country, author, and institution. Each level of co-authorship analysis provides insights into the nature of collaboration within the research community, highlighting the interconnectedness of research efforts across different dimensions. It identifies the core authors and the collaborative efforts

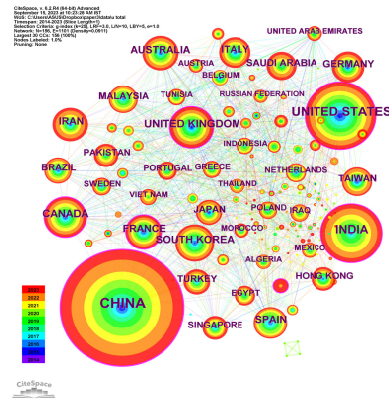


Fig. 4. Country co-authorship network.

among scholars, countries and institutions, thereby providing insight into the research development process. Across all three levels of co-authorship analysis, common network indicators such as node size, edge thickness, and BC play crucial roles in visualizing and interpreting the data. In these analyses, each node symbolizes an entity (such as a country, author, or institution), with the node's size reflecting the count of publications linked to that entity. A pink border surrounding a node signifies BC value, measuring how frequently the node appears on the shortest paths connecting other nodes. This metric is crucial for understanding the significance of specific entities within the network. The BC value is calculated using Eq. 2:

$$BC(node_v) = \sum_{s \neq v \neq t} \frac{\rho_{st}(v)}{\rho_{st}} \quad (2)$$

where, ρ_{st} represents the number of shortest paths between node s and t , and $\rho_{st}(v)$ is the number of those paths that pass through $node_v$. Links between nodes represent collaborative efforts, where thicker lines denote stronger collaboration based on the number of co-authored articles. The colored layers within a node indicate the period during which the publications were produced, with red highlighting more recent publications. These layers aid researchers in tracking the temporal progression of collaborations.

1) *Country*: Country co-authorship analysis focuses on understanding international research collaboration, revealing how countries partner on scientific projects. This analysis helps identify the global research networks and shows which countries play central roles in knowledge exchange and innovation. Fig. 4 shows the country co-authorship network diagram with 156 nodes, 1101 links, and 0.0911 network density, which indicates the closely connected collaborating countries in the ICT-assisted VRO research domain. Among the 156 countries in the network, the maximum are from Asia and Europe continents, with Asia having 38 countries and Europe following closely with 34 countries. This significant representation highlights the pivotal role both continents play in the global landscape of research and collaboration indicating a rich diversity of institutions and academic networks within these regions. Table III presents the rank-wise listing of the top 10 countries, detailing the number of publications (Frequency),

TABLE III
TOP TEN COLLABORATING COUNTRIES

Rank	Country	Frequency	Link Count	BC	Rank	Country	Frequency	Link Count	BC
1	China	2868	65	0.36	6	South Korea	327	49	0.05
2	United States	978	69	0.12	7	Australia	276	54	0.07
3	India	753	53	0.14	8	Iran	271	33	0.01
4	United Kingdom	386	64	0.11	9	France	264	52	0.10
5	Canada	360	58	0.12	10	Spain	257	51	0.09

neighboring nodes (Link Count), and BC in ICT-supported VRO literature. The analysis reveals that the United States demonstrates the highest degree of collaboration, with the most connections (69) in the country co-authorship network. Moreover, China (2868) is the leading contributor to ICT-assisted VRO literature produced in collaboration with other countries, followed by the United States (978), India (753), and the United Kingdom (386). From the context of BC, China (0.36), India (0.14), and the United States and Canada (0.12) are key collaborating countries. France possesses a lower publications count (264), but a higher BC value (0.10) which indicates that instead of fewer publications the country plays a key role in ICT-assisted VRO research. Moreover, among the top 10 countries, four are from Asia (China, South Korea, India, Iran), three are from Europe (United Kingdom, France, Spain), and two are from North America (United States and Canada).

2) *Authors*: Authors co-authorship analysis sheds light on the collaborative efforts among larger groups of researchers, driving research on the role of five ICT categories in VRO. The network in Fig. S1 represents the collaborative relationship among authors with five or more publications. The network has 504 nodes, 2656 edges and 0.021 network density, indicating closely connected network entities.

Additionally, the data of the top 10 authors in Table S1 presents the area of interest, publications count (Papers), institute (Affiliation) and country of the author, top three co-authors, total number of collaborating authors (Link count) and BC value. The results show that Wang Y and Li Y have maximum co-authors, 64 and 59 respectively, suggesting that their research is carried out in larger collaborative groups. Moreover, Wang Y frequently collaborates with researchers in AI, driving innovations in VRO models. Larger author teams enhance connectivity by creating multiple pathways for further collaboration. A researcher might start as a Ph.D. student under one of these key authors around Ph.D. programs and research grants, which are foundational mechanisms for collaboration. Moreover, larger teams enable more extensive cross-border research connections, as the collaboration dynamics are supported by joint efforts in projects such as international conferences, cross-institution Ph.D. training, and joint grant applications. Furthermore, ZHANG Y (0.07), WANG Y (0.06), LI Y (0.06) and LI X (0.06) with maximum BC value indicate that their collaborations are central to forming new research partnerships across institutions and countries. The statistics also reveal that most of the top collaborating authors are from China.

3) *Institutions*: Institution co-authorship analysis evaluates the collaborations between academies, universities, or research institutions. This provides insights into the research strength of each institution and its collaborative relationships. By

analyzing the interactions between institutions, we can uncover the most influential research centers, their connections to other institutions, and their role in driving scientific advancements. Fig. S2 represents the institution co-authorship network diagram of 203 institutions, publishing five or more articles in collaboration, on VRO in five ICT categories. Among these institutions, 103 are from Asia, 44 are from Europe, 30 are from North America, 10 are from Australia & Oceania, 10 are from the Middle East, and 3 are from Africa and South America. The visual representation of the network highlights the prominence of Beijing Jiaotong University, which stands as the largest node with 145 publications, indicating its central role in global collaboration followed by Southeast University & Xidian University (86), and Tsinghua University (78). Universities with high BC values, such as Beijing Jiaotong University (0.13) and Shanghai Jiao Tong University (0.10), act as key hubs, connecting otherwise distant institutions, and making them integral to the research network's cohesion and flow of knowledge.

Furthermore, Table S2 summarizes the top 10 institutions in terms of rank, area of interest, top three collaborating partners of the institution (Collaborating institution), number of papers published in collaboration (Papers), link count, and BC. The frequent appearance of Central South University as a collaborator across multiple universities emphasizes its importance in sustaining regional collaborations. A meticulous examination of the data reveals that all of these are universities except the Peng Cheng Laboratory which indicates that universities are the main driving force in the ICT-assisted VRO research domain. Moreover, all these institutions contributed 805 articles, which accounted for 11.34% of all articles (7097) in the current research domain. Most of these high-producing universities are located in Asia, specifically in China. In the current research domain, Beijing Jiaotong University has the highest number of collaborators (40), followed by Shanghai Jiao Tong University (38) and Tsinghua University (37), offering a collaborative research opportunity for the researchers in ICT-assisted VRO.

C. Geographical Distribution Analysis

The geographical distribution analysis showcases the contributions of various countries within a specific research area. Table IV lists the top five contributing countries along with their publication counts and respective shares within each category in ICT-assisted VRO research. Publications share corresponding to each category refers to the proportion of publications from each country relative to the total publications within that specific category. The results show that China, the United States, and the United Kingdom are the top three contributors to literature writing across all the categories.

TABLE IV
TOP FIVE PRODUCTIVE NATIONS IN EACH DOMAIN

ICT Category	Country1	Country2	Country3	Country4	Country5
CIT	China (274, 18.79%)	United States (141, 9.67%)	India (140, 9.60%)	France (108, 7.41%)	Iran (98, 6.72%)
AI	China (437, 39.33%)	United States (102, 9.18%)	India (93, 8.37%)	United Kingdom (72, 6.48%)	Iran (57, 5.13%)
LBS	China (375, 38.34%)	United States (146, 14.93%)	India (86, 8.79%)	Canada (44, 4.50%)	Iran (41, 4.19%)
PC	China (779, 44.82%)	United States (277, 15.94%)	India (219, 12.60%)	South Korea (92, 5.29%)	United Kingdom (79, 4.55%)
CT	China (558, 30.79%)	United States (258, 14.24%)	India (249, 13.74%)	South Korea (119, 6.57%)	United Kingdom (100, 5.52%)

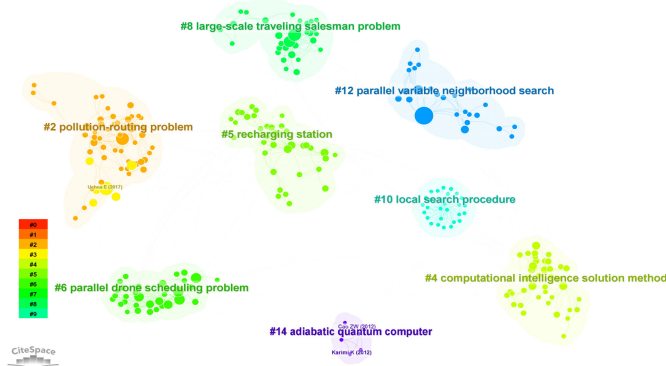


Fig. 5. Document co-citation network of CIT category.



Fig. 7. Document co-citation network of LBS category.

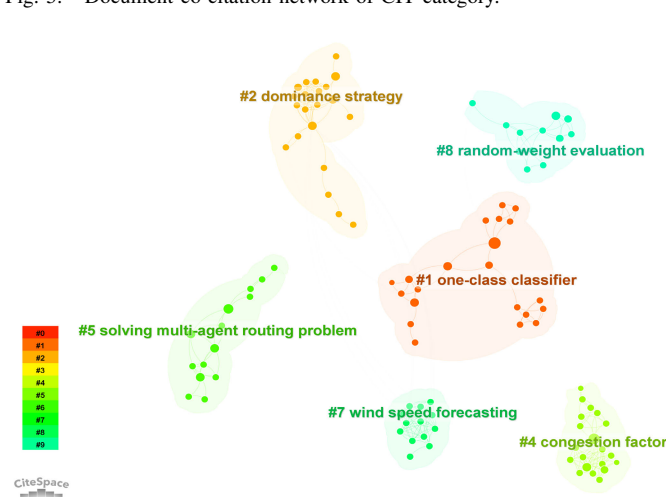


Fig. 6. Document co-citation network of AI category.

D. Document Co-Citation Network Analysis

In scientometrics, DCA is used to find relationships between references that are co-cited. This technique compiles the frequency of co-cited references and clusters them to identify evolutionary patterns and research themes. DCA has been applied in various research domains, including COVID-19 [10], quantum computing [31], [32], and airborne disease [33]. The CiteSpace tool can assist in visualizing references that are commonly cited in a specific subject domain and in identifying co-citation clusters. Henceforth, the authors employed CiteSpace to perform DCA and generate clusters corresponding to each category with the following parameters: (1) Time slice from 2014 to 2023; (2) Term source = title/abstract/author keywords/keywords plus; (3) Node type = reference; (4) Pruning = pathfinder/pruning the merged network; (5) Selection criteria g -index=25 (most cited articles per slice). The DCA networks are shown in Figs. 5, 6, 7, 8, 9. Different colors of the clusters in the network diagrams represent the size

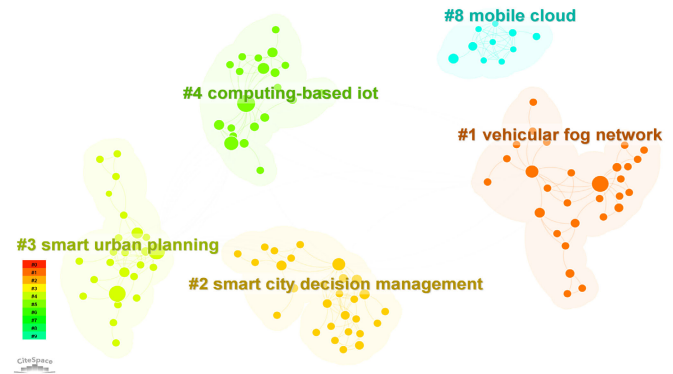


Fig. 8. Document co-citation network of PC category.

and significance of each cluster according to the scale. The red color indicates the largest and hence most significant cluster, while moving down the scale shows smaller and less central clusters in the network. Table V presents the attributes corresponding to the clusters in each category, including cluster ID, publications count (Size), Silhouette Score (SS), Mean Year (MY), cluster tag (LLR), and clusters linkage. The number followed by # in the network diagram represents the cluster ID specific to each category and does not correspond to another cluster ID across different categories. The phrase after the cluster ID is the cluster label. Cluster labels are drawn from index terms of citing articles, with top-ranked terms becoming the final cluster labels. These labels are extracted using three distinct algorithms latent semantic indexing [34], log-likelihood ratio (LLR) [35], and mutual information. The terms identified using LLR often highlight unique characteristics of the cluster, hence used in this analysis. The intellectual structure of a research domain is revealed through the analysis of how different works cite one another. Henceforth, the top citing articles within a cluster delineate the research front and improve the understanding of its intellectual base [24]. The SS value depicts the homogeneity of the cluster, values

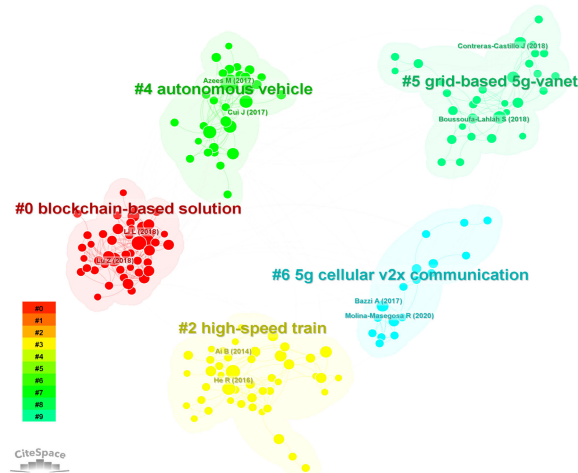


Fig. 9. Document co-citation network of CT category.

close to 1 indicate that articles within a cluster exhibit high consistency in content. The size of a cluster represents the number of articles it contains. Clusters with less than ten articles are excluded from the study as they are unlikely to significantly contribute meaningfully to the broader intellectual structure. The SS value of all network diagrams is greater than 0.8, indicating a high-quality cluster analysis of the ICT-based VRO. The MY of a cluster depicts the average publication year of the articles in that cluster. The coverage % presents the proportion of articles a citing article cites in that particular cluster, thereby providing a measure of the impact and influence of the work. The clusters linkage presents the thematic overlap and interdisciplinary connections between the five ICT categories to present a broader implication of the analysis.

Subsequently, the relevance of each cluster with VRO and the top three active citing articles corresponding to each cluster are discussed to delineate the intellectual structure of the current knowledge domain.

1) *Intellectual Structure of CIT*: In CIT, cluster #2 titled “pollution-routing problem”, is the largest cluster, containing 50 most cited articles published around 2011. The cluster includes various VRP variants focused on minimizing environmental impact through optimized routing strategies. Kramer et al. [36] presented a hybrid approach combining search-based metaheuristic with integer programming to minimize fuel consumption and environmental cost while taking capacity and time window constraints into consideration. Macrina et al. [37] presented an energy-efficient approach for green VRP with mixed vehicle fleets. Goeke and Schneider [38] proposed an adaptive large neighborhood search algorithm for the Electric Vehicle (EV) routing problem with time windows and mixed fleet for efficient load distribution and energy consumption.

Cluster #4 titled “computational intelligence solution method” contains 39 most cited articles published around 2015. This cluster applies AI-based techniques like game theory and Monte Carlo simulations to solve complex VRP scenarios. Mańdziuk [39] presented a survey on papers published in the field of VRP from 2015 to 2017, focusing

on new versions of VRP and pointed out that Monte Carlo simulations and game theory-based algorithms are in the current trend. Wang et al. [40] addressed the challenge of optimizing emergency transportation during real-life disasters by introducing a novel hybrid ant colony optimization-based algorithm. Feng et al. [41] presented a novel evolutionary multitasking algorithm for VRP with heterogeneous capacity, time window, and occasional drivers.

Cluster #5 titled “recharging station” contains 37 most cited articles published around 2019. The cluster is related to VRO as it optimizes routing for mixed fleets by incorporating charging strategies for electric vehicles. Wang and Zhao [42] proposed a path-based mixed-integer linear model and a hybrid heuristic combining large neighborhood search and set partitioning to address the electric fleet size and mix VRP with a partial linear recharging strategy. Amiri et al. [43] presented a mixed-integer linear programming model and developed two metaheuristic solutions for VRP with load capacity and time windows for a mixed fleet of EV and internal combustion vehicles, considering various charging technologies. Through experimentation in Scarborough, Ontario, Canada, the research highlights the need for EVs to have partial recharging and access to faster level 3 chargers when operating in urban settings. Keskin et al. [44] introduced a two-stage simulation-based heuristic that employs adaptive large neighborhood search to solve the EV routing problem, accounting for time windows and stochastic waiting times at charging stations.

Cluster #6 titled “parallel drone scheduling problem” contains 33 most cited articles published around 2018 which coordinate drones and trucks for efficient last-mile delivery and rural pickups. Saleu et al. [45] proposed a hybrid metaheuristic approach with mixed integer linear programming and branch and cut algorithms for solving VRP with drones to minimize the route completion time. Gunay-Sezer et al. [46] introduced a hybrid metaheuristic solution to solve Travelling Salesman Problem (TSP) with drones. Jiang et al. [47] introduced a multi-visit flexible-docking VRP that uses truck and fleet drones for pickup and delivery in rural areas.

Cluster #8 titled “large-scale traveling salesman problem” contains 30 most cited articles published around 2014 which extends classical TSP models to handle large-scale logistics and VRP applications. Jiang et al. [47] presented a novel pigeon-inspired optimization algorithm for solving large-scale TSP. Wang et al. [48] proposing a list-based simulated annealing algorithm for large-scale TSP is the second most citing article. Zhong et al. [49] presented a novel flight equation for particle swarm optimization algorithm to solve TSP.

Cluster #10 titled “local search procedure” contains 27 most cited articles published around 2009 which refine VRP solutions using iterative improvement methods for enhanced route optimization. Cui et al. [50] is the only citing article that cited all articles of this cluster. The authors proposed a quantum evolutionary algorithm with a local search procedure for solving capacitated VRP.

Cluster #12 titled “parallel variable neighborhood search” contains 25 most cited articles published around 2014 that employ parallel computing to tackle VRPs with uncertain

demand and time constraints. Hu et al. [51] proposed a novel framework with variable neighborhood search heuristic for VRP with hard time windows under demand and travel time uncertainty. Polat [52] and AbdAllah et al. [53], discussed divisible deliveries and pickups, and DVRP, respectively.

Cluster #14 titled “adiabatic quantum computer” contains 16 most cited articles published around 2013 which explore quantum algorithms for solving complex VRO problems faster and more efficiently than classical methods. As a most active citers Warren [54] with citing coverage of 60% and Cao and Elgart [55] presented the guiding path for executing VRP on adiabatic quantum computer.

2) *Intellectual Structure of AI*: In AI, cluster #1 titled “one class classifier” is the largest cluster containing 21 most cited articles published around 2017. One class classifier is used in VRO for anomaly detection in traffic data to improve vehicle routing efficiency. Yu et al. [56] developed an innovative neural combinatorial optimization approach based on deep reinforcement learning for real-time VRO with minimal completion time. Ferreira and Cortez [57] proposed a AutoOC to increase the predictive performance of one-class classifier. Mohanty et al. [58] presented an artificial chicken swarm algorithm for multi-objective optimization with DL.

Cluster #2 titled “dominance strategy” contains 21 most cited articles published around 2017. Dominance strategy is applied to optimize routes by evaluating and prioritizing traffic flow or path dominance, improving decision-making in routing. Abdel-Basset [59] proposed a hybrid whale optimization algorithm for multi-objective optimization, with citing coverage 57%. Ebadinezhad [60] enhanced ant colony optimization algorithm for TSP to address the slow convergence and local minima trapping problem. Wu et al. [61] presented a novel sparrow search algorithm for TSP.

Cluster #4 titled “congestion factor” contains 20 most cited articles published around 2016. The congestion factor is directly related to VRO by addressing traffic congestion, thus impacting the route planning. Zhang and You [62] developed a multi-population ant colony optimization algorithm to tackle large-scale TSP. Zhang et al. [63] proposed a multi-colony ant colony optimization for TSP. Gülcü et al. [64] hybrid ant colony optimization algorithm for TSP. Cluster #5 titled “solving multi-agent routing problem” contains 14 most cited articles published around 2018. The cluster focuses on optimizing routes for multiple vehicles simultaneously, which is critical in fleet management and logistics. Bono et al. [65] presented a sequential multi-agent decision-making model that adapts to changing conditions in DVRP. Xu et al. [66] provided an attention-based reinforcement learning for VRP that enhanced node embeddings. Zhao et al. [67] presented a novel deep reinforcement learning for VRP.

Cluster #7 titled “wind speed forecasting” contains 13 most cited articles published around 2017. Wind speed forecasting impacts route optimization in cases where wind conditions affect travel efficiency, particularly in logistics and transportation planning. He et al. [68] presented DL-based and Cheng and Wang [69] presented multi-objective based swarm intelligence optimization algorithms for wind speed forecasting.

Sanches et al. [70] proposed a quantum-inspired reinforcement learning-based approach for solving VRP.

Cluster #8 titled “random-weight evaluation” contains 12 most cited articles published around 2019. The cluster random-weight evaluation is a method used in ML approaches to evaluate and improve route optimization models, particularly in stochastic and uncertain environments. Bennett et al. [71] and Wang et al. [72] provided quantum-inspired solutions for solving VRP.

3) *Intellectual Structure of LBS*: In LBS, cluster #0 titled “wireless sensor network deployment” is the largest cluster containing 24 most cited articles published around 2016. The cluster is related to VRO in the context of using real-time data from distributed sensor networks to enhance traffic management and routing decisions. Bauzid et al. [73] proposed a genetic algorithm for optimal node placement in wireless sensor networks. Ahmed et al. [74] introduced a multi-objective whale optimization algorithm aimed at minimizing energy consumption and extending the lifespan of sink nodes in large-scale wireless sensor networks. Chauhan et al. [75] presented a multi-parent evolutionary algorithm designed for optimal head selection in wireless sensor networks.

Cluster #2 titled “uav-assisted wireless data” contains 16 most cited articles published around 2016. The cluster is related to VRO for leveraging Unmanned Aerial Vehicles (UAVs) to collect or transmit data that can support dynamic route adjustments in real-time. Zhang et al. [76] introduced a framework for multi-mode communication and optimal trajectory design within a radio network utilizing UAVs. Zeng et al. [77] proposed a trajectory design for minimum completion time in UAV-based multicasting. Zhan et al. [78] proposed an optimal trajectory design for UAV enabled wireless sensor network.

Cluster #3 titled “unpredictable environment” contains 15 most cited articles published around 2018. The cluster is relevant for addressing route optimization in scenarios with uncertainty or unpredictability, requiring adaptive algorithms to handle changing conditions. The three most active citers of this cluster articles Dell’Amico et al. [79], Vilorio [80], and Tamke and Buscher [81] proposed the role of drones in solving VRP.

Cluster #4 titled “medical applications” contains 14 most cited articles published around 2018. The cluster is relevant for emergency vehicle routing, where optimized paths are critical for timely medical deliveries or patient transport. The most active citing articles by Kumar and Kumar [82], Yu et al. [83] and Shen et al. [84], discussed various approaches for UAV path planning in emergency scenarios.

4) *Intellectual Structure of PC*: In PC, cluster #1 titled “vehicular fog network” is the largest cluster containing 29 most cited articles published around 2018. The cluster focuses on leveraging fog computing to process vehicle data locally, optimizing routes by reducing latency and improving decision-making at the edge. Liu et al. [85] presented an optimal task allocation approach in vehicular fog networks for intelligent vehicle transportation systems. Guzel and Ozdemir [86] presented an energy-efficient multi-objective optimization-based IoT service composition framework for

fog-based IoT networks. Zhou [87] presented a vehicular fog computing-based approach for resource management and task offloading.

Cluster #2 titled “smart city decision management” contains 28 most cited articles published around 2015. The cluster is related to VRO in integrating urban data and optimizing traffic flow and vehicle routing, to ensure efficient transport and resource management in smart cities. Ge et al. [88] performed a survey on big data technologies used in various IoT technologies. Silva et al. [89] proposed an architecture for real-time data processing aimed at urban planning and smart city decision management leveraging big data analytics. Ghofrani et al. [90] surveyed recent applications of big data analytics in railway transportation systems.

Cluster #3 titled “smart urban planning” contains 26 most cited articles published around 2014. The cluster literature uses urban data and simulation models to design cities and infrastructure that optimize traffic routing and vehicle flows. Ang et al. [91] performed a study on the deployment of the IoVs for smart cities. Lin et al. [92] performed a survey of IoT technologies from a fog/edge computing perspective. Silva et al. [93] presented an architecture for embedded smart city, based on big data analytics, to enhance real-time data processing and decision-making.

Cluster #4 titled “computing-based iot” contains 21 most cited articles published around 2017. The cluster focuses on utilizing IoT devices and cloud computing together, to optimize vehicle routes in real time by collecting and analyzing traffic data. Khalid et al. [94] proposed a fog-based framework for traffic light control in intelligent transportation systems. Sun et al. [95] presented an optical flow method approach for moving vehicle detection and tracking in complex transportation systems. Cui et al. [96] proposed a framework for optimal energy consumption in mobile edge computing for IoT.

Cluster #8 titled “mobile cloud” contains 10 most cited articles published around 2011. Mobile cloud computing helps manage and distribute traffic and route optimization tasks, allowing vehicles to compute optimal paths with remote support. Ghasemi et al. [97] and Zhou et al. [98] are the two most active citers in this cluster.

5) *Intellectual Structure of CT*: In CT, cluster #0 titled “blockchain-based solution” is the largest cluster containing 46 most cited articles published around 2018. The cluster is relevant for the secure and transparent management of data in VRO problem, ensuring tamper-proof routing decisions. Mikavica and Kostić-Ljubisavljević [99] performed a survey of blockchain-based solutions for security, privacy, and trust management in VANETs. Xu et al. [100] proposed a blockchain-based vehicle reputation management scheme for making the intelligent transportation system relying on IoV resistant to malicious attacks and selfish behaviors. Hussain [101] performed a survey on state-of-the-art secure communication in VANET.

Cluster #2 titled “high-speed train” contains 45 most cited articles published around 2014. The cluster focuses on optimizing routes for high-speed trains, contributing techniques that can inform VRO on land. Sun et al. [102] presented a framework for optimal dynamic resource management in

high-speed trains. Lu et al. [103] examined the impact of power adjustment on handover performance in high-speed railway communication systems to optimize green system design. Zhou et al. [104] explored the angular characteristics and spatial correlation of high-speed railway channels using a novel moving virtual antenna array measurement scheme.

Cluster #4 titled “autonomous vehicle” contains 31 most cited articles published around 2018. Autonomous vehicles are central to route optimization by enabling vehicles to independently determine optimal paths using AI and sensor data. Nayak et al. [105] performed a survey of VANETs from the perspective of resource allocation, security, and data privacy. Tomar and Tripathi [106] and Wang et al. [107] presented a blockchain-based certificateless authentication system for efficient communication in VANETs.

Cluster #5 titled “grid-based 5g-vanet” contains 31 most cited articles published around 2019. The cluster enlightens on to enhance VRO through efficient communication between vehicles and infrastructure, improving real-time route adjustments. Ullah et al. [108], Abbas et al. [109] and Kumbhar and Shin [110] proposed frameworks for emergency message dissemination in grid-based 5G-VANETs and high mobility networks.

Cluster #6 titled “5g cellular v2x communication” contains 17 most cited articles published around 2018. The cluster facilitates fast and reliable communication between vehicles and infrastructure, crucial for dynamic route planning. Ihsan et al. [111] proposed an energy-efficient power allocation scheme for multicasting beyond 5G V2X networks, using a low-complexity gradient-assisted binary search technique. Ullah et al. [112], [113] presented an energy-efficient resource allocation framework for 6G backscatter-enabled IoV networks. A meticulous analysis of the clusters shows some similarities and differences among them. Notably, the clusters #2, #6 in CIT; #5 in AI; #0 in LBS; #2, #3 in PC; #2, #4, #5 in CT shows that managing & optimizing transportation, data flow, and real-time decision making are recurring theme across all categories. Furthermore, advanced technologies including AI and IoT are integrated in multiple categories such as #1, #2 in AI; #0, #1 in LBS; #4 in PC; #6 in CT. Some clusters highlight the emergence of green and sustainable solutions, including #2, #5 in CIT; #7 in AI; #2, #3 in PC.

Moreover, the analysis also underscores significant differences among the clusters, with each category exhibiting unique domain-specific focuses for instance, clusters in PC are inclined towards addressing large-scale problems by #2, #3, #4; the clusters in CT focus on communication and its security by #0, #5, #6; clusters in AI focus on decision making and forecasting by #1, #2, #7, #8. Additionally, the clusters further differentiate themselves by delineating cutting-edge and niche technologies like adiabatic quantum computing, parallel drone scheduling, blockchain, 5G VANET, quantum-inspired reinforcement learning, and UAV-assisted data collection and path planning, while others focus on more established technologies such as IoT, fog computing, edge computing, big data analytics and various metaheuristic algorithms for solving variant of VRPs and DVRPs (such as problems with time window, heterogeneous capacity).

TABLE V
CLUSTERS SUMMARY OF ICT-ASSISTED VRO DOMAIN

Category	Cluster ID	Size	SS	MY	Cluster Tag (LLR)	Clusters Linkage
CIT	2	50	0.897	2011	pollution-routing problem	#5(CIT)
	4	39	0.879	2015	computational intelligence solution method	#5(AI)
	5	37	0.890	2019	recharging station	#2(CIT)
	6	33	0.948	2018	parallel drone scheduling problem	#8(CIT), #2(AI), #4(AI)
	8	30	0.897	2014	large-scale traveling salesman problem	#6(CIT), #2(AI), #4(AI)
	10	27	1	2009	local search procedure	#14(CIT), #8(AI)
	12	25	0.933	2014	parallel variable neighborhood search	#5(AI), #2(LBS), #2(CT), #6(CT)
	14	16	0.922	2020	adiabatic quantum computer one class classifier	#5(AI), #7(AI), #8(AI)
	1	21	0.993	2017	dominance strategy	#1(PC), #2(PC), #3(PC), #5(CT)
	2	21	0.916	2017	congestion factor	#6(CIT), #8(CIT), #4(AI)
AI	4	20	1	2016	solving multi-agent routing problem	#6(CIT), #8(CIT), #2(AI), #14(CIT), #2(LBS), #2(CT), #6(CT)
	7	13	0.98	2017	wind speed forecasting	#14(CIT), #8(AI)
	8	12	0.961	2019	random-weight evaluation	#10(CIT), #14(CIT), #7(AI)
	0	24	1	2016	wireless sensor network deployment	#4(PC)
LBS	2	16	0.958	2016	uav-assisted wireless data	#6(CIT), #3(LBS), #4(LBS)
	3	15	0.975	2018	unpredictable environment	#6(CIT), #2(LBS), #4(LBS)
	4	14	0.986	2018	medical application	#6(CIT), #2(LBS), #3(LBS)
	1	29	0.906	2018	vehicular fog network	#1(AI)
PC	2	28	0.968	2015	smart city decision management	#3(PC)
	3	26	0.979	2014	smart urban planning	#2(PC)
	4	21	0.943	2017	computing-based iot	#0(LBS)
	8	10	1	2011	mobile cloud	#0(CT), #4(CT)
CT	0	46	0.883	2018	blockchain-based solution	#4(CT)
	2	45	0.944	2014	high-speed train	#12(CIT), #5(AI), #2(LBS), #6(CT)
	4	31	0.867	2018	autonomous vehicle	#0(CT)
	5	31	0.904	2019	grid-based 5g-vanet	#6(CT)
	6	17	0.927	2018	5g cellular v2x communication	#12(CIT), #5(AI), #2(LBS), #2(CT)

Furthermore, the linkage between clusters has been presented in column “Clusters Linkage” of Table V to highlight the interdisciplinary connections and thematic overlaps. Specifically, clusters #2(CIT) and #5(CIT) are linked through a shared focus on EV routing problems with time windows and mixed fleet operations to optimize energy consumption. Similarly, cluster #1(AI) is linked with clusters #1(PC), #2(PC), #3(PC), #5(CT) by aligning the common objective of real-time VRO. Cluster #2(LBS) is linked with clusters #6(CIT), #3(LBS), #4(LBS) since they share the common theme of path planning and addressing TSP using drones or UAVs to collect or transmit network data for efficient VRO. Likewise, clusters #1(PC) and #1(AI) are linked for solving multi-objective optimization problems. Cluster #0(CT) is linked with clusters #12(CIT), #5(AI), #2(LBS), #2(CT) through DVRP theme. Cluster #7(AI) is linked with clusters #14(CIT) and #8(AI) for using quantum-based solutions to solve VRP. Similarly, other clusters also share common themes that contribute to the broader interdisciplinary connections within the field.

E. Top Articles by BC

Identifying top articles serves as a benchmark for excellence, guiding researchers toward high-quality research. BC metric in the DCA is indicative of potential pivotal points of paradigm shift over time. BC measure provides a computational method for finding influential and groundbreaking research within a field. Articles with strong BC scores are the intellectual work of the community [114]. Table VI presents the top five publications based on BC metric corresponding to each category of ICT in VRO. This section provides a detailed description of these articles and their relevant to the VRO research domain.

CIT: Reference [115] analyzed various heuristics and meta-heuristics for solving Multi-Attribute (MA) VRP, focusing

TABLE VI
TOP FIVE ARTICLES BY BC

Category	Reference	Title	BC
CIT	[115]	“Heuristics for multi-attribute vehicle routing problems: A survey and synthesis.”	0.13
	[116]	“Survey of green vehicle routing problem: past and future trends.”	0.11
	[117]	“An improved discrete bat algorithm for symmetric and asymmetric traveling salesman problems.”	0.11
AI	[118]	“Rich vehicle routing problems: From a taxonomy to a definition.”	0.11
	[135]	“The vehicle routing problem with drones: several worst-case results.”	0.11
	[119]	“Ant colony optimization with local search for dynamic traveling salesman problems.”	0.12
	[56]	“Online vehicle routing with neural combinatorial optimization and deep reinforcement learning.”	0.08
	[120]	“A hybrid ant colony optimization algorithm for a multi-objective vehicle routing problem with flexible time windows.”	0.08
	[121]	“Green logistic vehicle routing problem: Routing light delivery vehicles in urban areas using a neuro-fuzzy model.”	0.08
	[135]	“LSI-based simulated annealing algorithm for traveling salesman problem.”	0.08
	[122]	“Multi-objective optimization for distributed MIMO networks.”	0.20
	[123]	“Energy-efficient UAV communication with trajectory optimization.”	0.20
	[124]	“Energy efficient fault tolerant clustering and routing algorithms for wireless sensor networks.”	0.20
LBS	[125]	“A multi-objective and PSO based energy efficient path design for mobile sink in wireless sensor networks.”	0.20
	[136]	“Wireless sensor networks and multi-UAV systems for natural disaster management.”	0.20
	[126]	“Fog and IoT: An overview of research opportunities.”	0.19
PC	[127]	“Internet of things: A survey on enabling technologies, protocols, and applications.”	0.17
	[128]	“Minimizing cost and makespan for workflow scheduling in cloud using fuzzy dominance sort based HEFT.”	0.13
	[129]	“Multi-objective scheduling of extreme data scientific workflows in fog.”	0.09
CT	[137]	“Internet of things for smart cities.”	0.07
	[130]	“Multihop-cluster-based IEEE 802.11 p and LTE hybrid architecture for VANET safety message dissemination.”	0.19
	[131]	“Heterogeneous vehicular networking: A survey on architecture, challenges, and solutions.”	0.18
	[132]	“LTE for vehicular networking: A survey.”	0.17
	[133]	“Vehicular ad hoc networks (VANETs): status, results, and challenges.”	0.12
	[138]	“A comprehensive survey on vehicular ad hoc network.”	0.12

on real-world constraints and identified efficient methods from 64 high-performing meta-heuristics applied to different MAVRP variants. The work is relevant as it provides insights into designing better algorithms for optimizing complex vehicle routes in practical applications. Reference [116] reviewed the literature on Green VRP, classified Green VRP types, and identified research gaps related to environmental concerns in vehicle routing. The work is relevant as it explored how traditional VRP approaches could integrate sustainability, aiming to reduce both economic and environmental costs in logistics. Reference [117] proposed a discrete version of the bat algorithm to solve symmetric and asymmetric TSP and showed its effectiveness by comparing it against five optimization techniques across 37 instances. The work provides an enhanced metaheuristic approach for optimizing routes in various transportation and logistics scenarios. Reference [118] provided a comprehensive taxonomy, detailed definition, and analysis of various cases for Rich VRPs. This work aids in the development of more effective VRO solutions.

AI: Reference [119] presented a memetic ant colony optimization algorithm for solving DTSP, integrating a local search operator to enhance solution quality. This approach improves route optimization by adapting to changing travel times between nodes. The work is relevant to the VRO as it provides a robust method for optimizing routes in dynamic environments, applicable to real-world vehicle routing challenges. Reference [56] presented a deep reinforcement learning-based neural combinatorial optimization strategy for online vehicle routing, addressing the growing complexity of real-time transportation systems. By transforming the routing problem into a vehicle tour generation task, they use a graph-embedded pointer network to optimize routes efficiently. The proposed method is relevant to the VRO as it significantly reduces computation time, making it suitable for dynamic and large-scale logistic systems like last-mile deliveries. Reference [120] proposed an ant colony optimization approach with mutation operators and Pareto optimality for multi-objective VRP with flexible time windows, allowing vehicles to serve customers outside strict time frames within given tolerances, balancing cost savings and customer satisfaction. The proposed method

is relevant to the VRO as it effectively handles real-world logistics challenges by optimizing both cost efficiency and service quality, validated through benchmark tests and real-world analysis. Reference [121] proposed a model for routing light delivery vehicles that address environmental concerns such as exhaust emissions and noise in congested urban areas. The model considers both environmentally friendly and unfriendly vehicles, using an adaptive neural network trained by a simulated annealing algorithm to assess route performance based on logistics costs and environmental impact. The work incorporated environmental factors into the optimization process, providing a sustainable approach to urban logistics.

LBS: Reference [122] investigated three linear transceiver optimization in multiple-input multiple-output interference networks by addressing signal transmission, energy efficiency, and security using a multi-objective optimization problem-solving strategy. Reference [123] explored energy-efficient UAV communication by optimizing the UAV's trajectory to balance communication throughput and energy consumption. The proposed work is relevant to the VRO as it addresses the challenge of optimizing paths for energy efficiency, a common objective in vehicle routing scenarios. Reference [124] proposed a set of distributed clustering and routing algorithms for wireless sensor networks that address two key challenges: energy conservation and fault tolerance. The research is relevant to VRO as it highlights strategies for efficient routing and fault tolerance, crucial for optimizing vehicle routes under dynamic conditions. Reference [125] introduced a multi-objective particle swarm optimization algorithm aimed at optimizing the trajectory of mobile sink in wireless sensor networks to address the hot-spot or sink-hole problem by utilizing rendezvous point-based path design. This is relevant to VRO as it highlights techniques for optimizing routes under multiple objectives and constraints.

PC: Reference [126] presented a thorough survey on fog computing, and highlighted the architecture's versatility across mobile and wireline scenarios and its ability to support a wide range of applications, including IoT, 5G systems, and embedded AI. Its relevance to VRO lies in its application to IoT and 5G systems, which can enhance real-time decision-making and optimization in routing through improved data processing at the network edge. Reference [127] reviewed the technologies and protocols enabling the IoT, focusing on smart sensors and machine-to-machine communication. The integration of IoT with big data, cloud, and fog computing, and highlights key challenges. The insights are relevant to VRO through improved decision-making and system efficiency in intelligent transportation. Reference [128] addressed workflow scheduling in infrastructure as a service cloud platform by proposing a fuzzy-dominance-based algorithm to optimize both cost and makespan. Their solution, integrating fuzzy logic with traditional scheduling, is relevant to VRO as it similarly focuses on efficient resource allocation and optimization in complex environments. Reference [129] proposed a multi-objective workflow offloading for optimizing task offloading in fog computing, targeting extreme data workflows with strict latency, reliability, and cost constraints. This is relevant to VRO as both deal with optimizing routes/tasks

across distributed networks to minimize delays and maximize efficiency.

CT: Reference [130] proposed a hybrid architecture, a vehicular multihop algorithm for stable clustering with Long Term Evolution (LTE), combining IEEE 802.11p-based multihop clustering and the fourth-generation cellular system, to improve data delivery and reduce delays in VANETs. The work is relevant to VRO as both focus on optimizing communication and resource management across dynamic networks, ensuring efficient data transfer under varying conditions like vehicle density and mobility. Reference [131] surveyed recent advancements in heterogeneous VANETs, which integrate dedicated short-range communication and cellular networks to meet intelligent transportation system communication needs, especially for real-time safety services. The work relates to VRO by addressing the efficient coordination of communication networks. Reference [132] explored the potential of LTE for supporting vehicular applications, analyzing its strengths, limitations, and open research challenges compared to IEEE 802.11p. This is relevant to VRO as both involve optimizing systems—communication in this case—to enhance traffic efficiency, reliability, and safety in dynamic mobility scenarios. Reference [133] surveyed recent developments in VANETs, focusing on routing, broadcasting, quality of service, and security, while also reviewing wireless standards, trials, and simulation tools. The work aligns with VRO by addressing efficient routing and communication strategies within dynamic VANETs, essential for improving traffic efficiency and road safety.

These top articles by BC highlight influential research contributions across various categories of VRO within ICT. However, upon further examination, several articles demonstrate cross-topic integration, addressing more than one ICT category in relation to VRO. Notably, [56] leverages AI through deep reinforcement learning, and the use of a graph-embedded pointer network connects this work to LBS by optimizing vehicle tours based on spatial and temporal data. The relevance of the proposed method extends to real-time logistics, showcasing its applicability in large-scale systems like last-mile deliveries, which often involve elements of PC for real-time data processing and decision-making. This integration of AI, LBS, and PC underscores the multi-faceted nature of the solution. Reference [120] combines algorithms from AI with optimization techniques for logistics challenges, integrating AI with PC principles to solve VRPs with flexible time windows. Additionally, [121] incorporates CT by utilizing neural networks trained with adaptive algorithms, while also focusing on Green VRP principles to integrate sustainability in urban logistics. The work by [128] demonstrates cross-topic integration by combining CIT with PC through a fuzzy-dominance-based algorithm for workflow scheduling. These works underscore that solving complex VRO challenges often requires a holistic, cross-topic approach that leverages advancements across multiple ICT domains.

IV. DISCUSSION

The objective of the current research is to evaluate the current research landscape of scientific literature in ICT-assisted

VRO by addressing preset research questions using in-depth scientometric analysis of the literature from the past decade. Following are the findings presented as responses to the predefined research questions:

What are the annual publication trends in the ICT-assisted VRO research domain?

To address the first research question, this research provides an in-depth analysis of the yearly publication trends for each category. The analysis discerns that the publication rate in this knowledge domain has doubled in the last five years. Furthermore, a significant research impact and high-quality work are observed in the CT and PC categories.

Which are the collaboration patterns of authors, institutes and countries in the contemporary knowledge landscape of this field?

The co-authorship analysis performed to elucidate the collaboration patterns of countries, authors and institutions reveals that China and the United States are the most collaborating countries and China has emerged as the most significant country, acting as a key hub in collaborative efforts. Additionally, the analysis shows that Asia and Europe play a leading role in global collaboration networks within this research domain. Moreover, geographical distribution analysis outlines that China, the United States, and India are the most productive nations in individual publications. The analyses encourage policymakers and funding organizations to foster collaborations and international partnerships with these countries to drive economic growth and innovation in this knowledge domain. Authors co-authorship network analysis highlights the intricate collaborative dynamics within the research domain with Wang Y and Li Y emerging as key players, each having 64 and 59 co-authors, respectively, underscoring their involvement in large collaborative teams. Furthermore, authors like Zhang Y, Wang Y, Li Y, and Li X exhibit the highest BC value, signifying their pivotal role in facilitating new research partnerships across institutions and countries. Notably, the analysis shows that many top collaborating authors are based in China, positioning it as a hub for collaborative efforts in this research domain. The network fosters diverse research pathways, often rooted in Ph.D. programs, research grants, and joint projects across institutions, enhancing cross-border collaborations through mechanisms such as international conferences and cross-institutional training. Institutions co-authorship network analysis provides insights into the research strength, connectivity, and influence of key academic centers. Beijing Jiaotong University emerges as the most influential institution, with 145 publications and the highest BC (0.13), positioning it as a key hub for global knowledge exchange. Universities such as Tsinghua University, Dalian University of Technology, and Shanghai Jiao Tong University also play critical roles in linking otherwise distant institutions and facilitating knowledge flow. Central South University frequently appears as a collaborator, emphasizing its importance in fostering regional partnerships.

What are the prevailing research themes and various ICTs effectively used in the last decade?

The study through DCA identifies that EV-based sustainable VRO, UAV-based wireless sensor network, drones-based VRO,

intelligent vehicle transportation system, V2X, and VANET are current intellectual research themes. Moreover, quantum computing, adiabatic quantum computing, hybrid ant colony optimization, and blockchain have emerged as the most popular technologies for addressing VRO problems.

Which are the significant publications that have shaped the field of ICT-assisted VRO.

This paper summarizes key articles to serve as the foundation for an in-depth understanding of recent trends, and technologies and to discern future research directions for academic professionals. Specifically, references from [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], [112], [113] extracted by conducting a comprehensive DCA corresponding to each category, are considered highly active citing articles in their respective clusters. Additionally, articles from [56], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136], [137], [138] are identified as significant articles based on BC value in the domain.

What are the future research directions, algorithm selection and research implications pertinent to the current state of knowledge in ICT-assisted VRO?

Future Research Directions:

- 1) Insights from clusters #14(CIT), #5 and #7(AI), #4(PC), and #4(CT), suggest that emerging solution methodologies, such as the application of the DL method, quantum computing, chaos theory, and reinforcement learning, have yet to gain widespread popularity within the VRO research community. There exists a substantial opportunity for researchers to explore the integration of quantum computing with classical algorithms and ML. This hybrid approach has the potential to exponentially speed up complex routing scenarios, such as multi-objective optimization and large-scale VRO problems, with unprecedented efficiency, thus reducing fuel consumption and greenhouse gas emissions.
- 2) From clusters #2(LBS), #1(PC), and #5(CT), the blending of emerging technologies such as IoV, and edge computing frameworks with UAVs for smart cities VRO to solve distributed system-related issues, emerge as an intriguing avenue for exploration.
- 3) Findings from the clusters #0 and #4(CT), highlight the incorporation of blockchain technology to address data privacy and security concerns is becoming increasingly vital, particularly given the growing reliance on real-time tracking and data-sharing mechanisms.
- 4) From clusters #3(LBS), #2 and #3(PC), it is discerned that interdisciplinary research that combines ICT with logistics, urban planning, and environmental sciences can unlock new avenues for efficient, sustainable, and resilient VRO in the coming years.

- 5) Insights from clusters #2, #5 and #12 (CIT), suggest that in EV-enabled energy-efficient and sustainable VRO, optimizing charging infrastructure placement for EV fleets and developing dynamic routing algorithms to minimize energy consumption is a new research area for exploration. Addressing these limitations promises to make substantial strides toward more sustainable transportation systems.

Algorithm Selection for VRO:

Recently, the maturity of exact and metaheuristic methods for tackling large-scale VRO problems has been observed. Nevertheless, the researchers are now taking a progressive stride by exploring innovative hybrid methodologies that seamlessly blend exact and metaheuristic approaches. A careful analysis of key articles discerns that the range of hybrid (hybrid ant colony optimization, hybrid heuristic combining large neighborhood search and set partitioning), metaheuristic and heuristic (adaptive large neighborhood search algorithm, monte carlo simulations, game theory-based algorithms, evolutionary multitasking algorithm, pigeon-inspired optimization algorithm, simulated annealing algorithm, variable neighborhood search heuristic, memetic ant colony optimization algorithm, various swarm intelligence algorithms), quantum algorithms (quantum-inspired reinforcement learning, quantum evolutionary algorithm, adiabatic quantum computing), drone & uav scheduling algorithms (parallel drone scheduling algorithms, trajectory design algorithms), machine learning approaches (fuzzy-dominance-based algorithm, deep reinforcement learning-based neural combinatorial optimization strategy, attention-based reinforcement learning, deep reinforcement learning) contribute to advancing the field of VRO by addressing diverse constraints, uncertainties, and real-time decision-making challenges.

Further analysis outlines when to use different algorithms in VRO:

- 1) Hybrid methods, adaptive large neighborhood search, mixed-integer linear programming, and simulation-based heuristics are used for solving green and energy-efficient EV routing with time windows and recharging station management to minimize environmental impacts and sustainable transportation [38], [43], [44].
- 2) Monte Carlo simulations, game-theory based algorithm, pigeon-inspired optimization, simulated annealing, variable neighborhood search, quantum evolutionary algorithm, and adiabatic quantum computing are used for solving capacity-constrained and multi-objective VRP, large-scale TSP. Moreover, adiabatic quantum computing explores VRP execution on quantum computers for faster solutions, guiding research toward practical applications of quantum algorithms in logistics [39], [47], [48], [51], [54], [55], [121].
- 3) Hybrid metaheuristics such as genetic, branch and cut, whale optimization, and evolutionary algorithms are used for solving parallel drone scheduling and UAV path planning [41], [50], [59], [73], [74].
- 4) Various AI methods including one-class classifier, deep reinforcement learning and quantum-inspired reinforcement learning algorithms are used for real-time routing scenarios such as online vehicle routing with anomaly detection, dynamic VRP, multi-agent routing, and wind speed forecasting [56], [57], [67], [70].
- 5) Genetic, multi-parent evolutionary, multi-objective swarm intelligence and distributed clustering algorithms are used for solving optimizing node placement in wireless sensor networks, UAV-assisted data collection for dynamic routing, energy-efficient routing in sensor networks with fault tolerance and route optimization in emergency scenarios [69], [75], [124].
- 6) Blockchain-based solutions and low-complexity binary search techniques are used for secure routing and V2X communication, respectively [99], [100], [107], [111].
- 7) Fog/edge computing-based approaches are used for enhancing decision-making at the network edge, reducing latency in vehicle routing [85], [86], [87], [92], [94], [126], [127], [129].

Research Implications:

- 1) The current research provides a comprehensive overview of ICT in the VRO domain, aiding the transport researcher community to quickly capture the current state of research, the collaboration trends, the understanding of its intellectual base, and current challenges.
- 2) The co-authorship analyses foster knowledge exchange, accelerate innovation, enhance research quality and promote comprehensive solutions for intermodal transport challenges.
- 3) The intellectual structure of the domain helps the practitioners for keeping them abreast of the latest developments in the field, thus formulating the best strategies, and opting best models for applying their efforts in the most efficient way.
- 4) Findings of the present article on the foremost cutting-edge adiabatic quantum computing technology, hybrid ant colony optimization, UAV-based wireless sensor network, IoV, and blockchain help the software engineers and model developers in adopting the significant/popular technique for achieving outstanding results.
- 5) The articles with high BC help the operational research team to uncover hidden trends and connect fragmented knowledge across fields for dynamic problem-solving and real-time decision-making in the transport infrastructure.

V. CONCLUSION AND RESEARCH LIMITATIONS

The current article serves as a comprehensive representation of the current research progress in ICT-assisted VRO. The scientometric analysis is performed on the basis of five major categories, viz., Computational Intelligence Techniques (CIT), Artificial Intelligence (AI), Location Based Services (LBS), Pervasive Computing (PC), Communication Technologies (CT), offering valuable insights and visual maps of VRO-related literature that aids researchers and software practitioners in comprehending the research landscape, collaborating community, and intellectual base. Moreover, a critical literature review of active citing articles and top articles based on BC is conducted to provide future research directions, research implications and to discern specific research

strategies, algorithms, tools, or frameworks used by individual papers.

The research identifies a rapid increase in ICT-assisted VRO publications, with output rate doubling over the last five years and notable contributions in the CT and PC categories. Collaboration networks reveal China and the United States as dominant players, with Asia leading in global research partnerships, supported by institutions like Beijing Jiaotong University. Research themes focus on sustainable VRP using electric vehicles, drones, and VANETs, alongside emerging technologies such as quantum computing, hybrid metaheuristics, and blockchain. Future directions emphasize integrating quantum algorithms, IoV, and blockchain with classical optimization techniques, with interdisciplinary research driving innovation in sustainable and resilient transport systems. This review presents valuable articles for the researchers and comprehensive insights for experienced practitioners, emphasizing the significance and requirement of advancing innovation and practical applications in ICT-assisted VRO.

The current research work has the following research limitations.

- 1) The study provides insights into important trends and techniques identified through the scientometric analysis, but it does not attempt to solve these challenges. Future research can build on these findings to develop or improve practical solutions for the critical aspects identified, such as gaps in research or under-explored areas.
- 2) Although the work summarizes full-text articles in DCA and top articles based on BC to discern specific research strategies, algorithms, tools, or frameworks used by individual papers, but it focuses on performing the higher-level bibliometric analysis using research documents' metadata. Including full-text articles can provide an avenue for future research to reveal intent for deeper insights.
- 3) This article focuses on identifying and analyzing recent research trends within the domain, specifically considering studies published after 2014. By excluding earlier works, the study emphasizes contemporary advancements and emerging themes. However, it provides an opportunity for future researchers to explore seminal and foundational studies predating 2014, to examine the historical development and evolution of the field.

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